Hardik .K. Singh

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Education

Vellore Institute of Technology, Oct 2022 – July 2026
B.Tech in Computer Science and Engineering Specialization in Health Informatics **(GPA: 7.2/10.0)**

Skills

Data Analysis: Python, SQL, Pandas, NumPy, Data Cleaning, EDA.
ML/AI: TensorFlow, Computer Vision, OpenCV.
Web Development: React.js, Java, HTML/CSS, JavaScript.
Tools: Git, Power BI, VS Code.

Projects

Healthcare Data Analysis Dashboard
Python, SQL, Pandas, Matplotlib, Seaborn, Power BI

- Extracted, cleaned, and analyzed 10,000+ patient records from relational databases using SQL and Python,performing validation, outlier handling, and EDA to uncover disease trends and hospital KPIs, improving data quality by 40
- Conducted exploratory data analysis and feature engineering to identify key patterns in medical imaging data, optimizing data preprocessing pipelines and creating standardized datasets that enhanced ML model performance and reduced training time by 30%.

Healthcare Management System
React.js, Node.js, Express.js, MongoDB, REST API, HTML/CSS, JavaScript, Git

- Engineered a full-stack MERN healthcare management application featuring patient registration, doctor profile management, appointment scheduling, and secure user authentication, with RESTful API endpoints built using Node.js and Express.js and MongoDB as the scalable, cloud-hosted database.
- Designed and implemented a responsive React.js frontend using component-based architecture and mobile-first UI principles, integrating JWT-based authentication and real-time appointment management to deliver a seamless end-to-end user experience with 99% uptime on Render/Vercel deployment.

Disease Prediction AI Model
Python, Scikit-learn, Pandas, NumPy, Flask

- Trained ML classification models (Logistic Regression, Random Forest, Decision Tree) on symptom-based datasets, with feature engineering and hyperparameter tuning to reach 92%+ accuracy across 40+ disease classes.
- Built and deployed a Flask web app for real-time disease prediction from user symptoms, optimizing the inference pipeline with Scikit-learn and NumPy to reduce latency by 45% while keeping high accuracy.

Certifications

- MERN Full Stack Development Certification –Ethnus
- AI Foundation Course –Jio Institute

Awards & Recognition

Health Hackathon Participant | Johns Hopkins University, USA

- Selected among top participants to develop AI-powered healthcare solutions addressing real-world medical.

Second Position, IEEE Q-RiOSITY Competition | VIT Bhopal

- Secured 2nd rank among 100+ participants demonstrating engineering knowledge and rapid problem-solving.